Ch. 06. Numerical Differentiation

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Finite Difference Method

- Let us suppose that we are looking for the derivative of a function f(x) at some given point x.
- Assume that the function f(x) is known at equally spaced point x_i, such that $h=x_{i+1} - x_i$ is the spacing between nodes. Let

$$f_i = f(x_i) \qquad \text{for} \quad i = 0, \ldots, N_x - 1$$

- In order to find the derivative $f' = df/dx$, the most direct method expands the function using a Taylor series in the neighborhood of x_i:

$$f_{i+1} \equiv f(x_i + h) \approx f_i + f_i' h + \frac{f_i''}{2} h^2 + \frac{f_i'''}{3!} h^3 + O(h^4)$$

- Solving for f'_i, we have the *forward difference (FD)* approximation:

$$f_i' \approx \frac{f_{i+1} - f_i}{h}$$

- This approximation has an error $\varepsilon = f'' h/2$: we can make the approximation error smaller by making *h* smaller, yet precision will be lost through the subtractive cancellation on the left-hand side when h is too small.

Backward Difference

- Similarly, we could expand $f(x_i-h)$:

$$f_{i-1} \equiv f(x_i - h) \approx f_i - f_i' h + \frac{f_i''}{2} h^2 - \frac{f_i'''}{3!} h^3 + O(h^4)$$

and obtain the *backward difference (BD)* approximation

$$f_i' \approx \frac{f_i - f_{i-1}}{h}$$

which still has the same error $\varepsilon = f'' h/2 = O(h)$.

- Both the forward and backward approximations are only first-order accurate and would give the correct answer only when $f(x)$ is a linear function.
- For a quadratic function $f(x)=a+bx^2$, for instance, the forward derivative approximation would result in

$$\frac{f_{i+1} - f_i}{h} = 2bx_i + bh$$

- If you compare it with the exact derivative ($f' = 2bx$), this clearly becomes a good approximation only for small h ($h << 2x_i$)

Central Difference

- Now consider both the right and left expansions:

$$\begin{cases} f_{i+1} \approx f_i + f_i' h + \dfrac{f_i''}{2} h^2 + \dfrac{f_i'''}{3!} h^3 + O(h^4) \\ f_{i-1} \approx f_i - f_i' h + \dfrac{f_i''}{2} h^2 - \dfrac{f_i'''}{3!} h^3 + O(h^4) \end{cases}$$

- Subtracting the two equations yields the *central difference (CD)* approximation

$$f_i' = \frac{f_{i+1} - f_{i-1}}{2h}$$

- During the subtraction, even powers cancel and our approximation is thus second-order accurate. The error is $\varepsilon = f''' h^2/6 = O(h^2)$ and you can expect the cd approximation to be exact for a parabola.

- The FD, BD and CD approximations are quite natural in the sense that they are reminiscent of the incremental ratio used in elementary calculus.

Higher Order Formulas

- It is possible to obtain higher-order approximation by including more points.

- If we now expand also f_{i+2} and f_{i-2}, we obtain a system of equations

$$\begin{cases} f_{i+2} \approx f_i + 2f_i'h + \frac{f_i''}{2}(2h)^2 + \frac{f_i'''}{3!}(2h)^3 + O(h^4) \\ f_{i+1} \approx f_i + f_i'h + \frac{f_i''}{2}h^2 + \frac{f_i'''}{3!}h^3 + O(h^4) \\ f_{i-1} \approx f_i - f_i'h + \frac{f_i''}{2}h^2 - \frac{f_i'''}{3!}h^3 + O(h^4) \\ f_{i-2} \approx f_i - 2f_i'h + \frac{f_i''}{2}(2h)^2 - \frac{f_i'''}{3!}(2h)^3 + O(h^4) \end{cases}$$

- Getting rid of terms up the fourth derivative, we obtain

$$f_i' \approx \frac{f_{i-2} - 8f_{i-1} + 8f_{i+1} - f_{i+2}}{12h}$$

which is a fourth-order accurate approximation with error $\varepsilon = f^{(5)} h^4/30$.

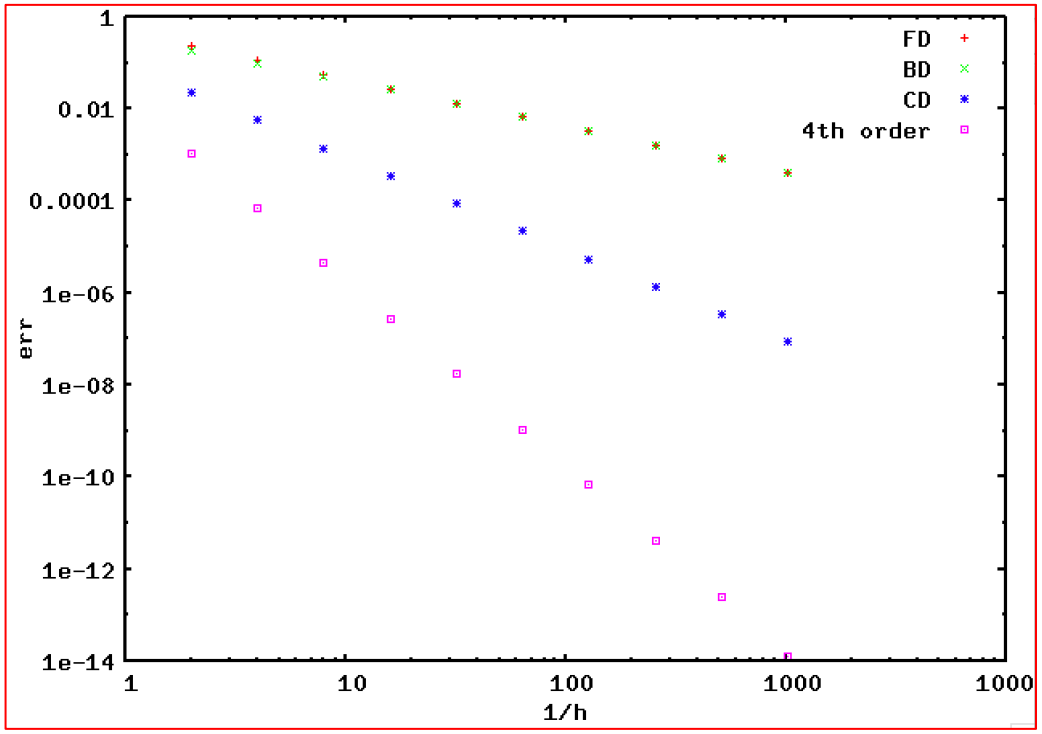
Practice Session #1

- **`derivative.cpp`**: compute the numerical derivative *`f(x)=sin(x)`* in *`x=1`* using FD, BD and CD (or higher) using different increments *`h=0.5,0.25,0.125,`* …

  Plot the error

$$\epsilon = \left| f'_{\text{num}} - f'_{\text{ex}} \right|$$

as a function of h using a log-log scaling.



2nd- and Higher-order Derivatives

- For higher order derivatives we can still make use of the Taylor expansion and solve for the second (or higher) derivative.

- From

$$\begin{cases} f_{i+1} \approx f_i + f_i' h + \frac{f_i''}{2} h^2 + \frac{f_i'''}{3!} h^3 + O(h^4) \\ f_{i-1} \approx f_i - f_i' h + \frac{f_i''}{2} h^2 - \frac{f_i'''}{3!} h^3 + O(h^4) \end{cases}$$

we can solve, e.g., for the 2nd derivative:

$$f_i'' \approx \frac{f_{i+1} - 2f_i + f_{i-1}}{h^2} + O(h^2)$$

Practice Session #2

- **`trajectory.cpp`**: Given the particle trajectory

$$x(t) = \alpha t^2 - t^3 \left[1 - exp\left(-\frac{\alpha^2}{t}\right)\right]$$

produce a plot of the velocity and acceleration in the range $0<t<\alpha$.
How many inversion points are present ? (try $\alpha=10$ to begin with)

To this purpose, divide the range `[0, α]` into *N* equally spaced intervals Δt and use this spacing when computing the derivatives (that is, h = Δt).

Note: x(t) has different L/R limits at t=0. Thus, central 1st and 2nd derivatives are ill-defined. At this point, replace them with the corresponding one-sided approximation.

